

HDFS

Sure! Let's talk about the Hadoop Distributed File System (HDFS) in simple terms.

HDFS is like a big library where instead of books, we store huge amounts of data. Imagine you have a giant book that is too big to fit on one shelf. HDFS helps us by breaking that big book into smaller chapters (called blocks) and placing those chapters on different shelves (or computers) in the library. This way, if one shelf gets damaged, we can still find the chapters on other shelves, making our library very safe and reliable.

Another cool thing about HDFS is that it allows us to read the chapters many times without changing them. It's like being able to borrow a book from the library, read it, and then return it without ever writing in it. If you want to add new information, you can append it to the end, but you can't change what's already there.

HDFS

- HDFS is the acronym for Hadoop Distributed File System
- It is the storage layer of Hadoop
- Splits the files into blocks, creates replicas of the blocks, and stores them on different machines
- Provides access to streaming data
- HDFS uses a command line interface to interact with Hadoop



Streaming means that HDFS provides a constant bitrate when transferring data rather than having the data being transferred in waves

- HDFS stands for Hadoop Distributed File System. A distributed file system is a file system that is distributed on multiple file servers and allows programmers to access or store files from any network or computer. HDFS is the storage layer of Hadoop. HDFS works by splitting the files into blocks, then creating

replicas of the blocks, and storing them on different machines. HDFS is built to access streaming data seamlessly. Streaming means that HDFS provides a constant bitrate when transferring data rather than having the data being transferred in waves. HDFS uses a command line interface to interact with Hadoop.

Key features

Cost efficient	The storage hardware is not expensive
Large amounts of data	HDFS can store up to petabytes of data
Replication	Makes copies of the data on multiple machines
Fault tolerant	If one machine crashes, a copy of the data can be found somewhere else and work continues
Scalable	One cluster can be scaled into hundreds of nodes
Portable	Can easily move across multiple platforms

- The key features of HDFS are: The commodity hardware that stores the data is not expensive, and therefore reduces storage costs. HDFS can store very large amounts of data, as large as petabytes— in any format, tabular and non-tabular. It splits these large amounts of data into small chunks called blocks.
- One of the great features of HDFS is its ability to replicate and minimize the costs associated with data losses when there is failure with one of the hardware units. That capability makes HDFS fault tolerant. In the event of a data loss of one of the computers, the data can be found on another computer and work continues.
- HDFS is also highly scalable. A single cluster can scale into hundreds of nodes. Portability is also one of the key features, as HDFS is designed to easily move from one platform to another.

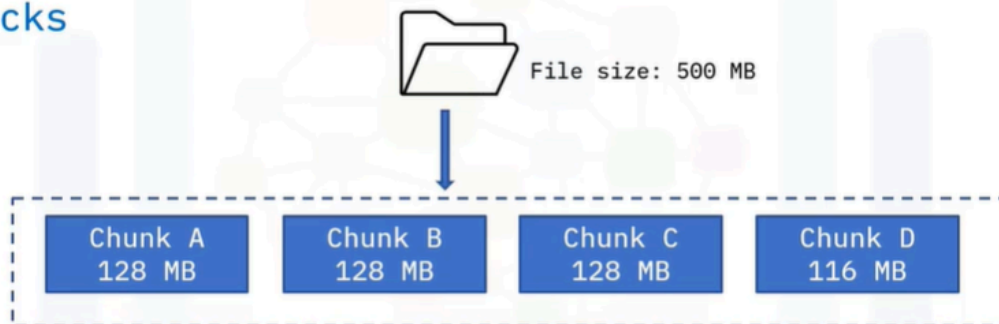
HDFS concepts

Blocks

- Minimum amount of data that can be read or written
- Provides fault tolerance
- Default size is 64 MB or 128 MB

HDFS concepts

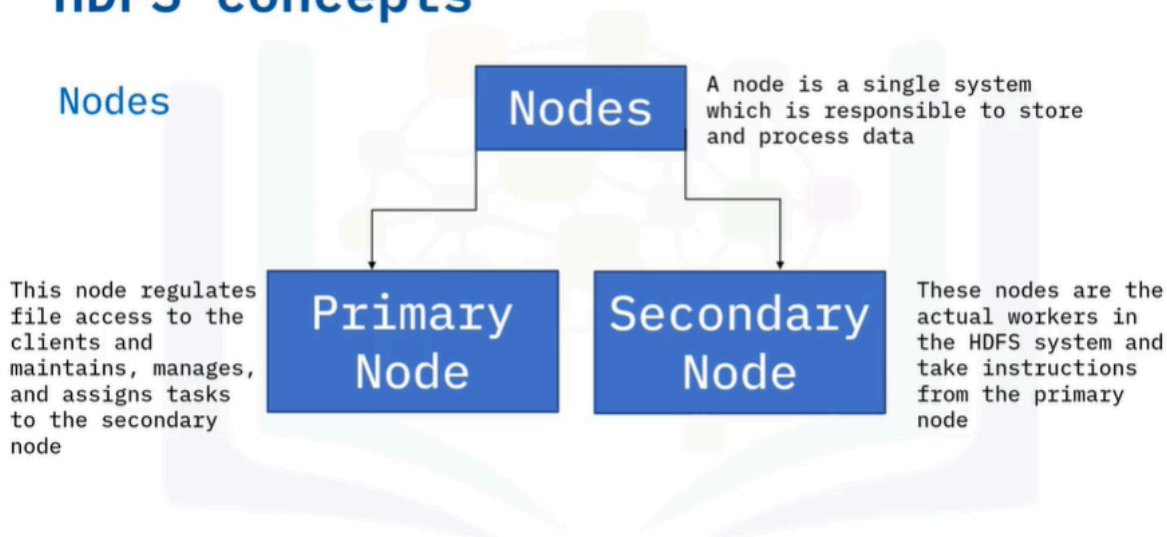
Blocks



- Each file stored doesn't have to take up the configured space size
- To get familiar with Hadoop, there are a few concepts to know. When HDFS receives files, files are broken into smaller chunks called blocks. A block is the minimum amount of data that can be read or written and provides fault tolerance. Depending on your system configuration, the default block size could be 64 or 128 megabytes. For example, if we had a 500-megabyte file, with a default block chunk size of 128 megabytes, the file will be divided into 3

blocks of 128 megabytes and one block of 116 megabytes. The only time you will have equal splits is if the file size is a multiple of the default block size. Therefore, you can see that each file stored doesn't have to take up the storage of the pre-configured block size.

HDFS concepts



- Let's start with nodes. A node is a single system which is responsible for storing and processing data. Think about it as one machine or computer in which data is stored. Remember that HDFS follows the primary/secondary concept. HDFS has two types of nodes:
 - The Primary node, known as the name node, regulates file access to the clients and maintains, manages, and assigns tasks to the secondary node, also known as a data node. There can be hundreds of data nodes in the HDFS that manage the storage system.
- They perform read and write requests at the instruction of the name node.

HDFS concepts

Rack awareness in HDFS

- Choosing data node racks that are closest to each other
- Improves cluster performance by reducing network traffic
- Name node keeps the rack ID information
- Replication can be done through rack awareness



A rack is the collection of about 40 to 50 data nodes using the same network switch

- When performing operations like read and write, it is important that the name node maximizes performance by choosing the data nodes closest to themselves. This could be by choosing data nodes on the same rack or in nearby racks.
- This is called rack awareness. A rack is the collection of about forty to fifty data nodes using the same network switch. Rack awareness is used to reduce the network traffic and improve cluster performance. To achieve rack awareness, the name node keeps the rack ID information. Replication is done by rack awareness as well. It is done by making sure replicas of a data node are in different racks. So, if a rack is down, you can still obtain the data from another rack.

HDFS concepts

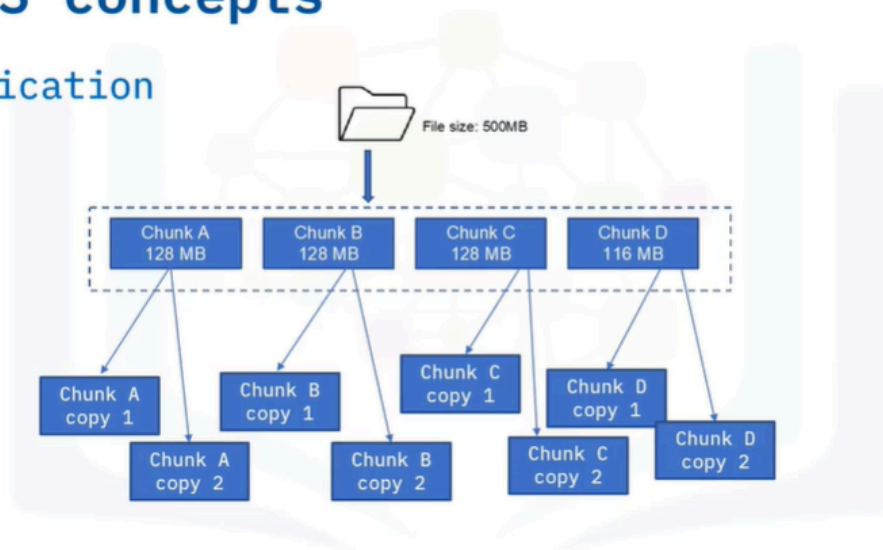
Replication

- Creating a copy of the data block
- Copies are created for backup purposes
- Replication factor: Number of times the data block was copied

- HDFS is known for optimizing replication. HDFS uses the rack awareness concept to create replicas to make sure that the data is reliable and available, and that the network bandwidth is properly utilized. Replication is creating a copy of the data block. When crashes happen, replication provides backup of the data blocks. Replication factor is defined as the number of times you make a copy of the data block. Depending on your configuration, you can set the number of copies you want.

HDFS concepts

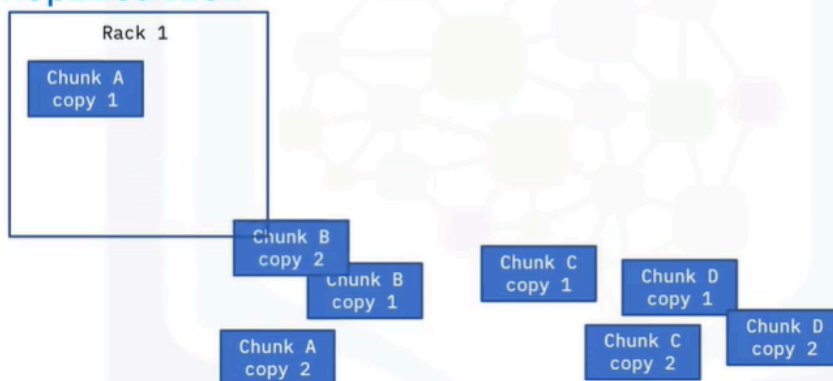
Replication



- Let us look at our 500-megabyte file example. If our replication factor is 2, it will create two copies of each block. That means we will have 4 times 2, which is eight copies of the file blocks for backup.

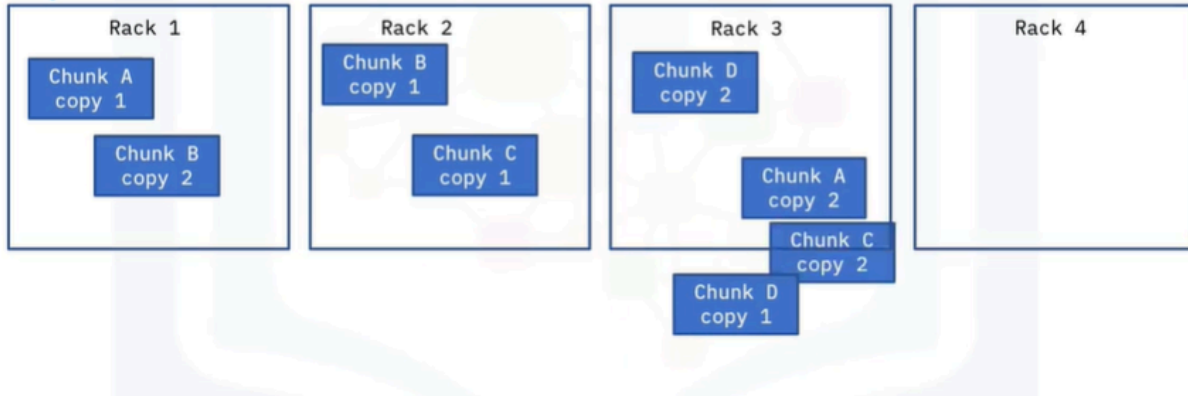
HDFS concepts

Replication



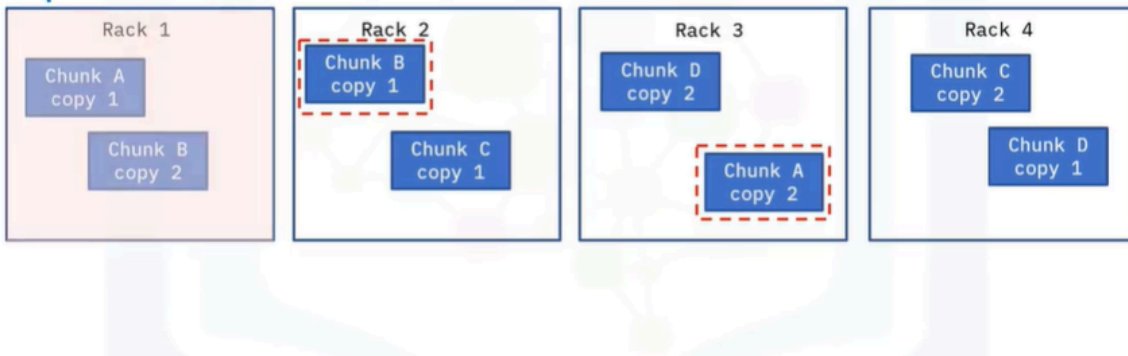
HDFS concepts

Replication



HDFS concepts

Replication



- HDFS uses the rack awareness concept and saves the blocks in different racks to make sure that a copy is available in another rack. For example, you could have four racks of two copies each. If one of the racks is down or crashes, you will still have a copy of the data blocks in a different rack, and you can still work with the data.

HDFS concepts

Read and Write Operations

- HDFS allows **write once read many** operations

Read

- Client will send a request to the primary node to get the location of the data nodes containing blocks



A client fulfills a user's request by interacting with the Name node and Data nodes

HDFS concepts

Read and Write Operations

- HDFS allows **write once read many** operations

Read

- Client will send a request to the primary node to get the location of the data nodes containing blocks
- Client will read files closest to the data nodes

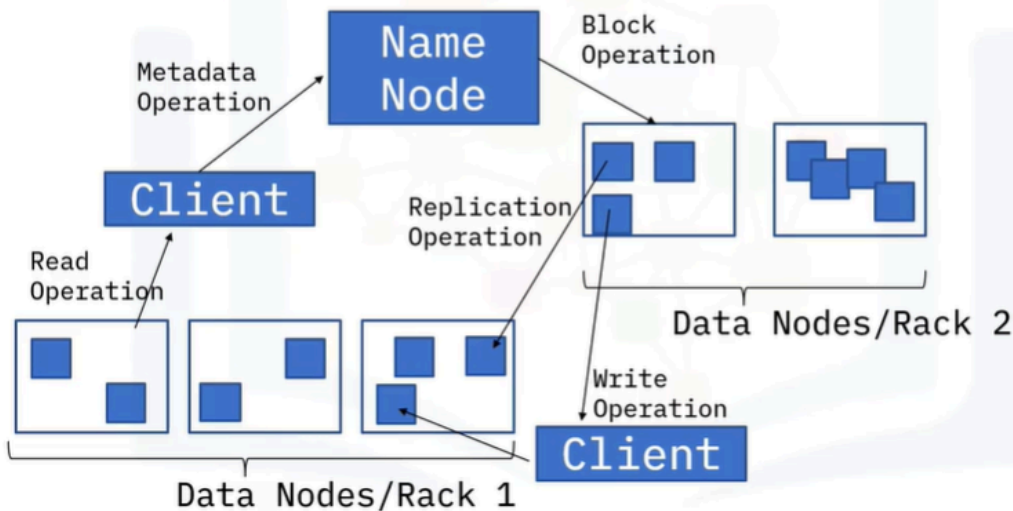
Write

- The Name node makes sure that the file doesn't exist
- If the file exists client gets an **IO Exception** message
- If the file doesn't exist, the client is given access to start writing files

- Another important concept is how read and write operations are performed in HDFS. HDFS allows “write once, read many” operations. This means that you cannot edit files that are already stored in HDFS, but you can append new data to them. Let us start with how the read operation works. Assuming we have a text file, the client will send a request to the primary node, which is the name node, to get the location of the data nodes containing blocks. The name node will verify that the client has the correct privileges and provide the client with the locations.

- A client in HDFS interacts with the primary and secondary nodes to fulfill a user's request. The client will then send a request to the closest data nodes through an FS Data Input stream object by calling the read method to read all the files. And when the client is done, the client will use the close method to end the session.
- Next, we will see how the write operation works. Just like in the read operation, the name node confirms that the client has the right privileges. The name node makes sure to check that the file doesn't exist in the system. If the file already exists, the client will receive an IO exception. If the file doesn't exist, the client receives a write permission together with the data nodes. Once the client is done, the data nodes start creating replicas and sends a confirmation to the client

HDFS architecture



- Hadoop follows the concept of a primary/secondary node architecture. The primary node is the name node. The architecture is such that per cluster, there is one name node and multiple data nodes, which are the secondary nodes. Internally, a file is split into one or more blocks and these blocks are stored in a set of data nodes.

- The name node oversees opening, closing, renaming file operations, and mapping file blocks to the data node. The data nodes are responsible for read and write requests from the client and perform the creation, replication, and deletion of file blocks based on instructions from the name node.

Summary

In this video, you learned that:

- Key HDFS benefits include its cost efficiency, scalability, data storage expansion and data replication capabilities
- A block is the minimum size of a file
- Replication is used for backup purposes
- Rack awareness helps reduce the network traffic and improve cluster performance
- HDFS enables **write once read many** operations